



```
In [1]: import pandas as pd
import matplotlib as mptb
import seaborn as snb
```

```
In [2]: titanic=pd.read_csv(r"C:\Users\Xservice5433\Downloads\tested.csv")
titanic.head(11)
```

```
Out[2]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	892	0	3	Kelly, Mr. James	male	34.5	0	0	33091
1	893	1	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	36327
2	894	0	2	Myles, Mr. Thomas Francis	male	62.0	0	0	24027
3	895	0	3	Wirz, Mr. Albert	male	27.0	0	0	31515
4	896	1	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	310129
5	897	0	3	Svensson, Mr. Johan Cervin	male	14.0	0	0	753
6	898	1	3	Connolly, Miss. Kate	female	30.0	0	0	33097
7	899	0	2	Caldwell, Mr. Albert Francis	male	26.0	1	1	24873
8	900	1	3	Abraham, Mrs. Joseph (Sophie Halaut Easu)	female	18.0	0	0	265
9	901	0	3	Davies, Mr. John Samuel	male	21.0	2	0	A/ 4887
10	902	0	3	Ilieff, Mr. Ylio	male	NaN	0	0	34922

```
In [3]: titanic.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     418 non-null    int64
1   Survived        418 non-null    int64
2   Pclass          418 non-null    int64
3   Name            418 non-null    object
4   Sex             418 non-null    object
5   Age            332 non-null    float64
6   SibSp           418 non-null    int64
7   Parch           418 non-null    int64
8   Ticket          418 non-null    object
9   Fare           417 non-null    float64
10  Cabin           91 non-null     object
11  Embarked        418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB

```

```
In [4]: titanic.isnull().sum()
```

```

Out[4]: PassengerId     0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                 86
SibSp                0
Parch                0
Ticket               0
Fare                 1
Cabin              327
Embarked             0
dtype: int64

```

```
In [6]: titanic['Age'] = titanic['Age'].fillna(titanic['Age'].median())
```

```
In [7]: titanic['Cabin'] = titanic['Cabin'].fillna('Unknown')
```

```
In [8]: titanic.isnull().sum()
```

```
Out[8]: PassengerId    0
        Survived      0
        Pclass        0
        Name          0
        Sex           0
        Age           0
        SibSp         0
        Parch         0
        Ticket        0
        Fare          1
        Cabin         0
        Embarked      0
        dtype: int64
```

```
In [9]: titanic['Fare'] = titanic['Fare'].fillna(titanic['Fare'].median())
```

```
In [10]: titanic.isnull().sum()
```

```
Out[10]: PassengerId    0
        Survived      0
        Pclass        0
        Name          0
        Sex           0
        Age           0
        SibSp         0
        Parch         0
        Ticket        0
        Fare          0
        Cabin         0
        Embarked      0
        dtype: int64
```

```
In [11]: titanic.describe()
```

```
Out[11]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	418.000000	418.000000	418.000000	418.000000	418.000000	418.000000
mean	1100.500000	0.363636	2.265550	29.599282	0.447368	0.392344
std	120.810458	0.481622	0.841838	12.703770	0.896760	0.981429
min	892.000000	0.000000	1.000000	0.170000	0.000000	0.000000
25%	996.250000	0.000000	1.000000	23.000000	0.000000	0.000000
50%	1100.500000	0.000000	3.000000	27.000000	0.000000	0.000000
75%	1204.750000	1.000000	3.000000	35.750000	1.000000	0.000000
max	1309.000000	1.000000	3.000000	76.000000	8.000000	9.000000

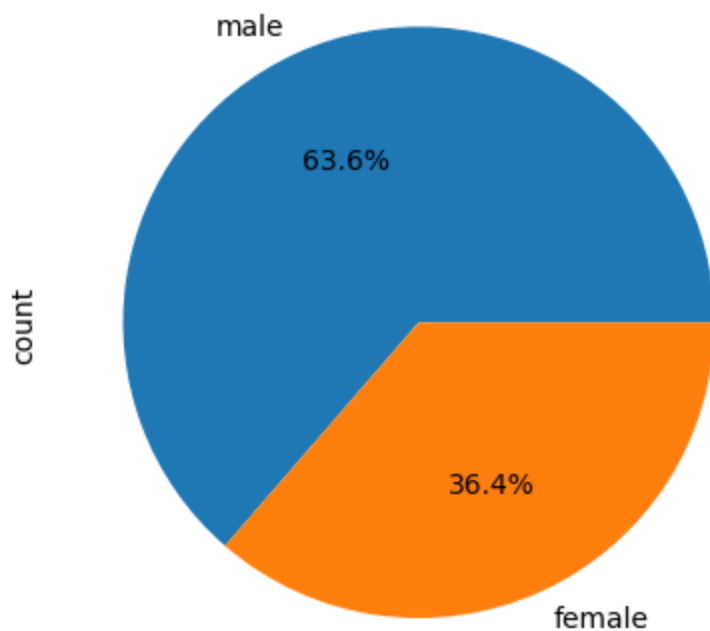
```
In [12]: titanic.value_counts()
```

```
Out[12]: PassengerId  Survived  Pclass  Name
Sex      Age      SibSp  Parch  Ticket      Fare      Cabin  Embarked
892      0      3      Kelly, Mr. James
male    34.5    0      0      330911      7.8292    Unknown  Q
1
893      1      3      Wilkes, Mrs. James (Ellen Needs)
female  47.0    1      0      363272      7.0000    Unknown  S
1
894      0      2      Myles, Mr. Thomas Francis
male    62.0    0      0      240276      9.6875    Unknown  Q
1
895      0      3      Wirz, Mr. Albert
male    27.0    0      0      315154      8.6625    Unknown  S
1
896      1      3      Hirvonen, Mrs. Alexander (Helga E Lindqvist)
female  22.0    1      1      3101298     12.2875    Unknown  S
1

..
1305     0      3      Spector, Mr. Woolf
male    27.0    0      0      A.5. 3236      8.0500    Unknown  S
1
1306     1      1      Oliva y Ocana, Dona. Fermina
female  39.0    0      0      PC 17758      108.9000    C105     C
1
1307     0      3      Saether, Mr. Simon Sivertsen
male    38.5    0      0      SOTON/O.Q. 3101262  7.2500    Unknown  S
1
1308     0      3      Ware, Mr. Frederick
male    27.0    0      0      359309      8.0500    Unknown  S
1
1309     0      3      Peter, Master. Michael J
male    27.0    1      1      2668      22.3583    Unknown  C
1
Name: count, Length: 418, dtype: int64
```

```
In [20]: # Q1. Gender wise Ticket Sold?
import matplotlib.pyplot as plt
titanic['Sex'].value_counts().plot.pie(autopct="%1.1f%%")
plt.title('Percentage of Sex wise Ticket Sold', fontsize=16)
plt.show()
```

Percentage of Sex wise Ticket Sold



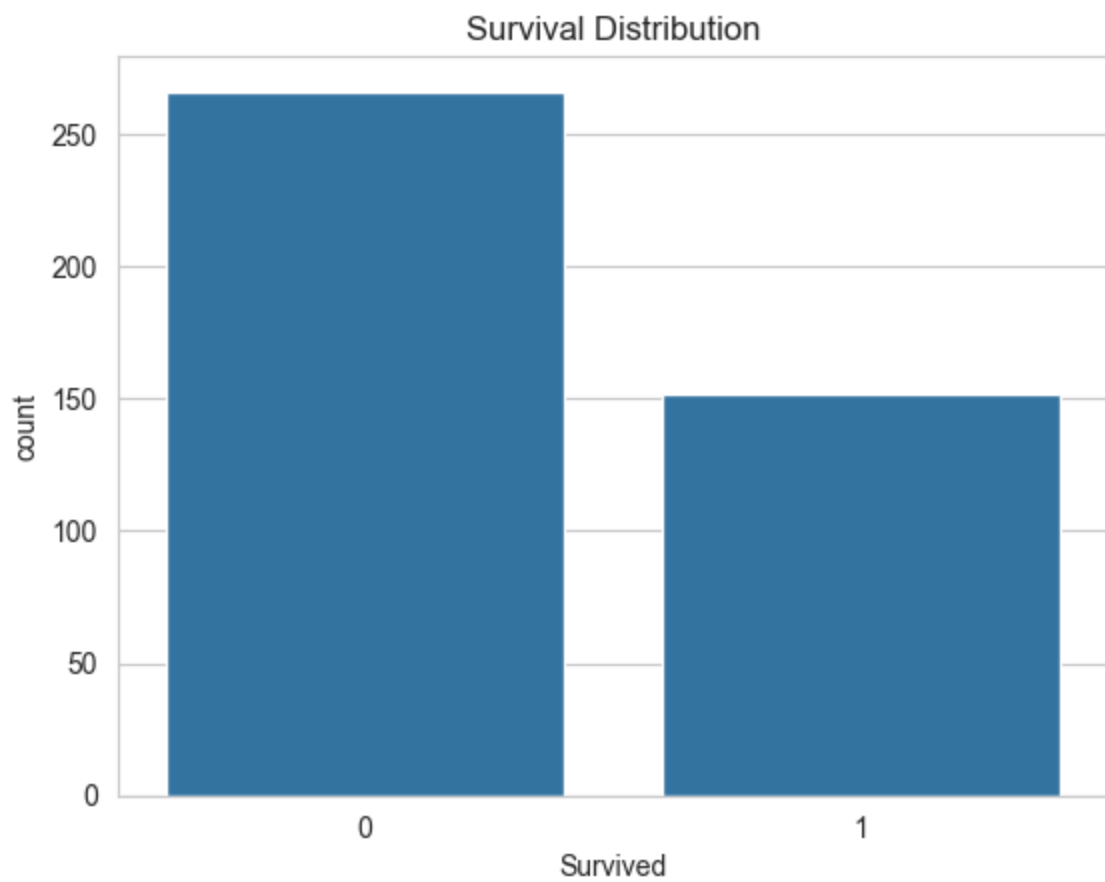
```
In [22]: #Q2. What is the overall survival rate?
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

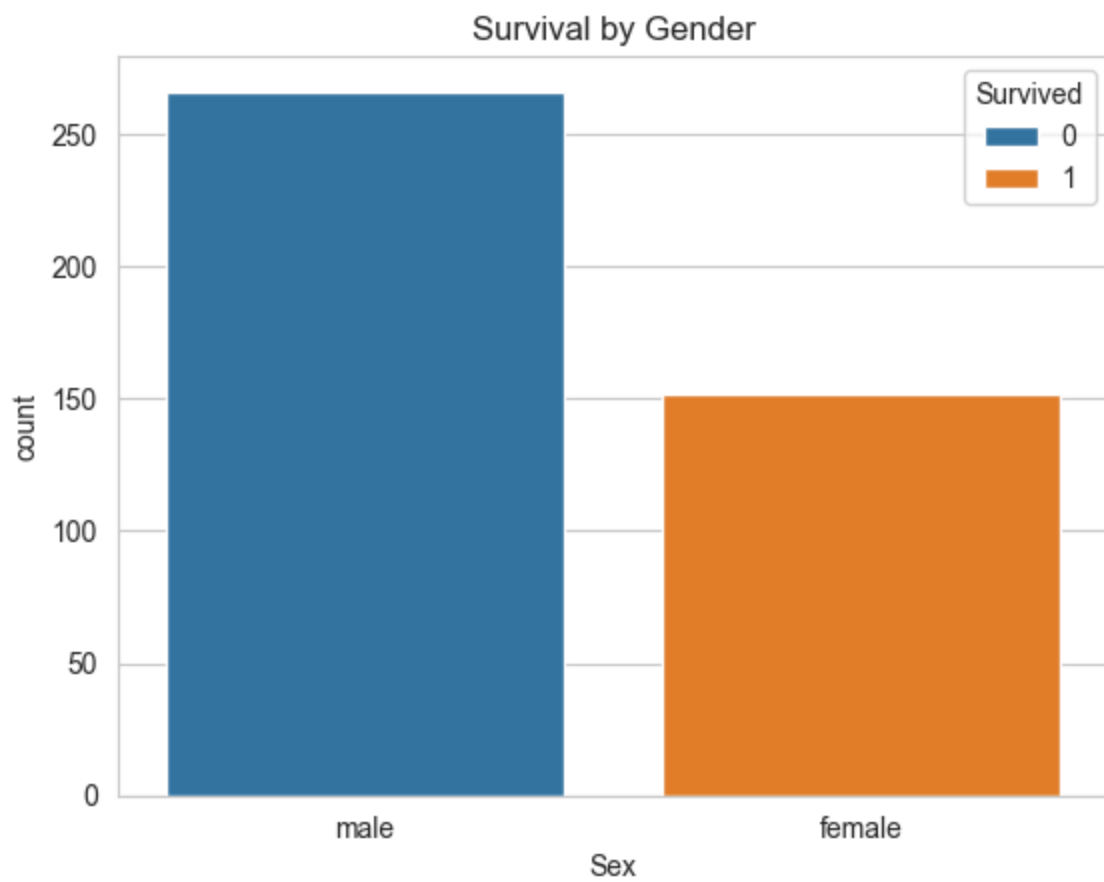
# Survival rate
survival_rate = titanic['Survived'].mean() * 100
print(f"Overall Survival Rate: {survival_rate:.2f}%")

sns.countplot(x='Survived', data=titanic)
plt.title("Survival Distribution")
plt.show()
```

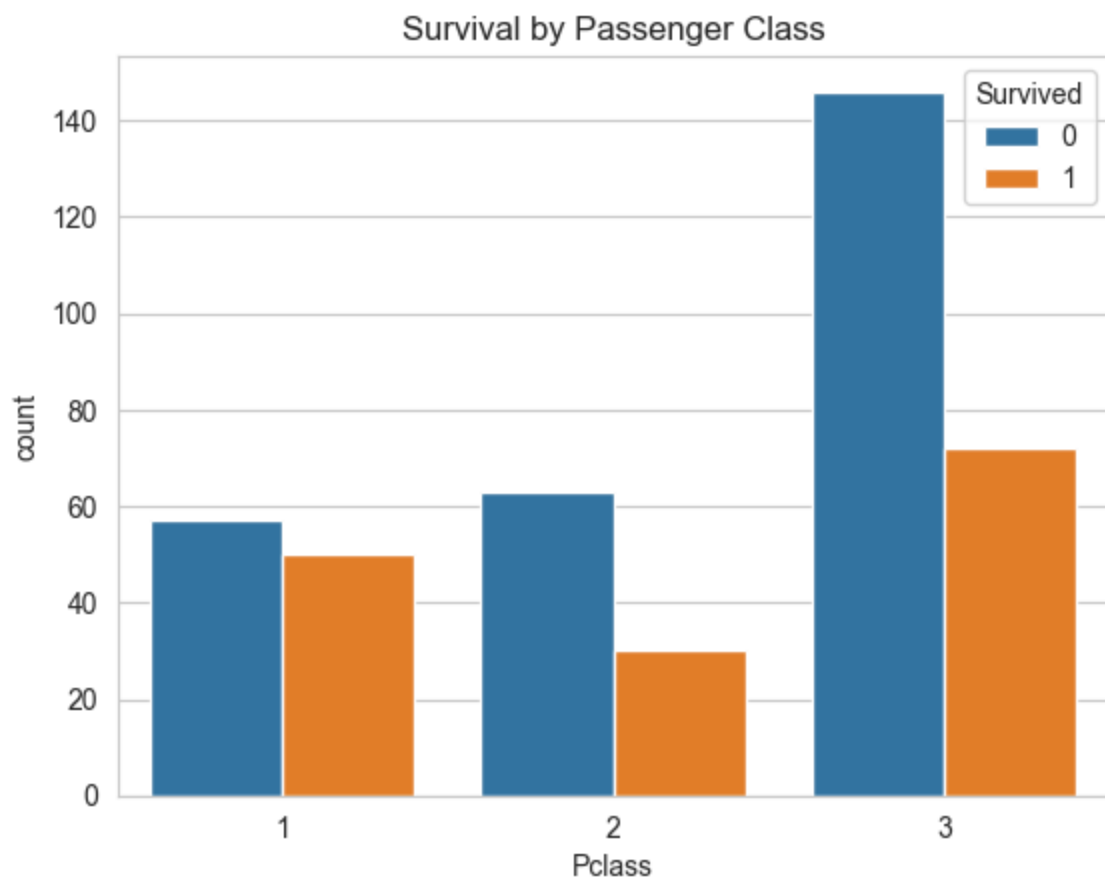
Overall Survival Rate: 36.36%



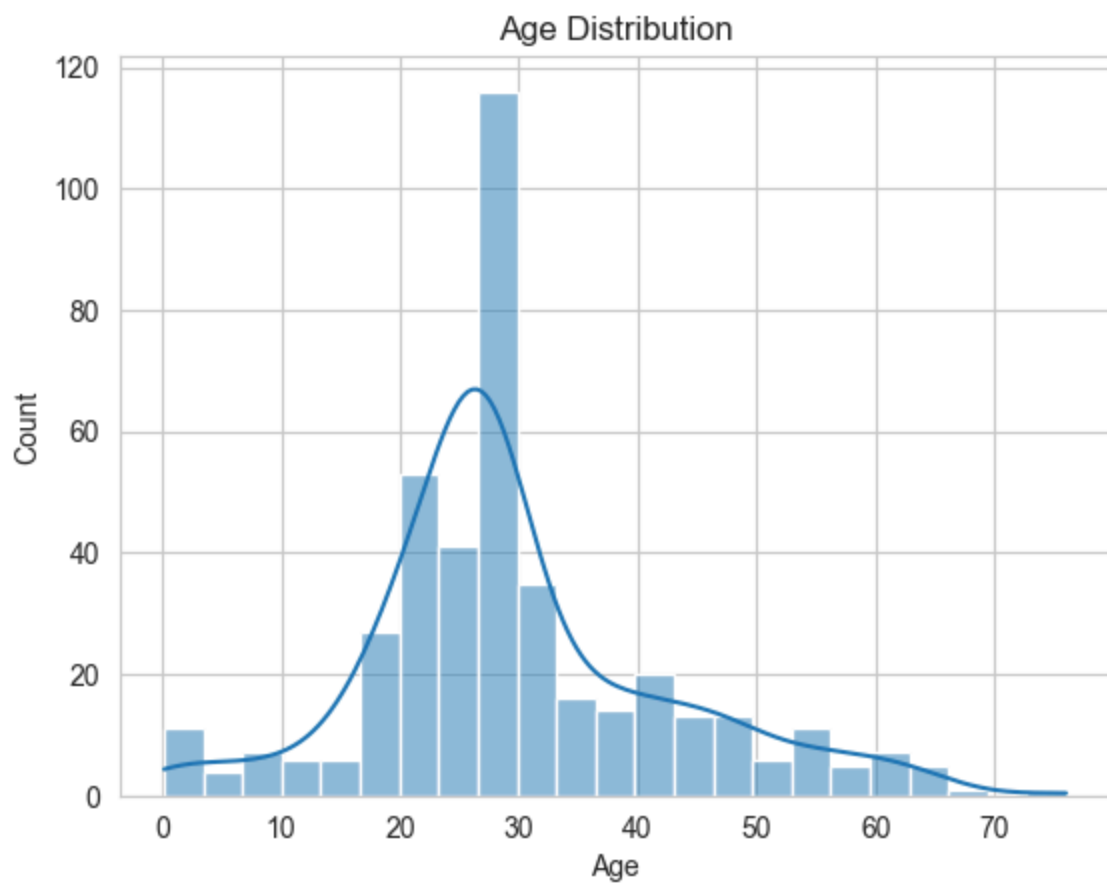
```
In [23]: #Q3. Does Gender Impact Survival?
sns.countplot(x='Sex', hue='Survived', data=titanic)
plt.title("Survival by Gender")
plt.show()
```



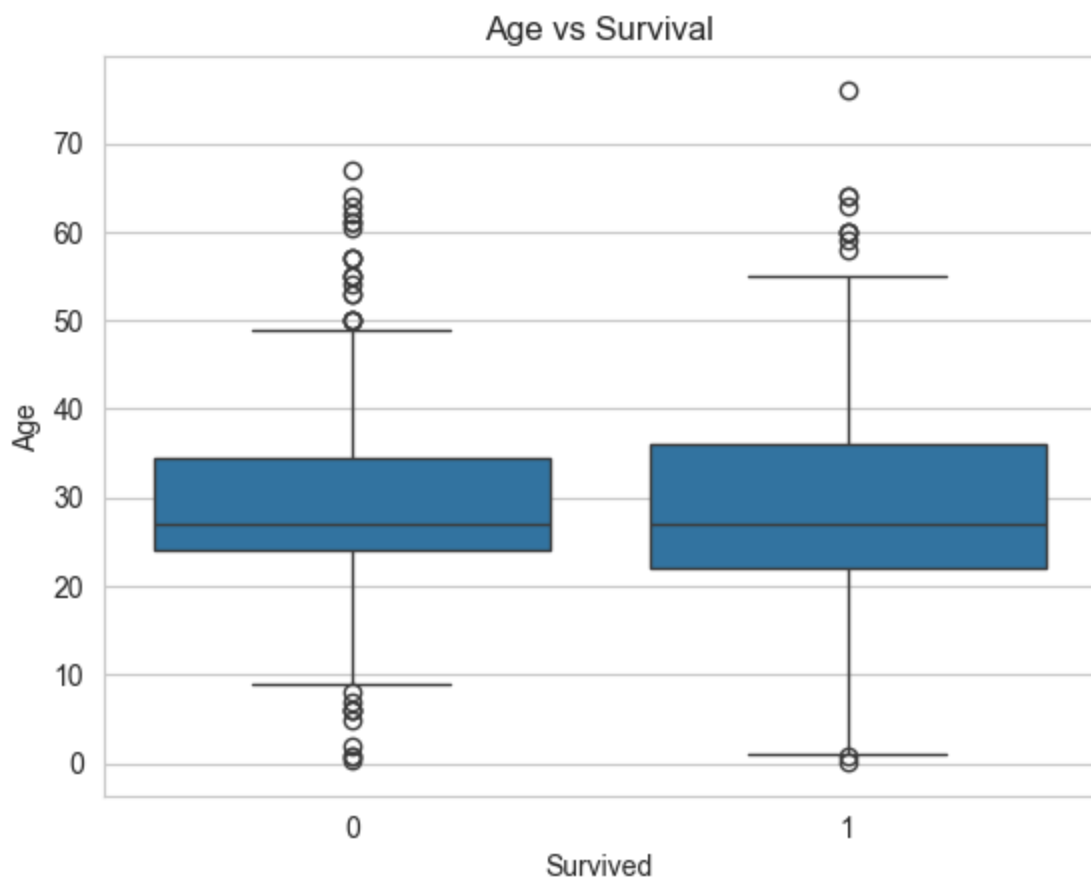
```
In [24]: #Q4. Does Passenger Class Affect Survival?
sns.countplot(x='Pclass', hue='Survived', data=titanic)
plt.title("Survival by Passenger Class")
plt.show()
```



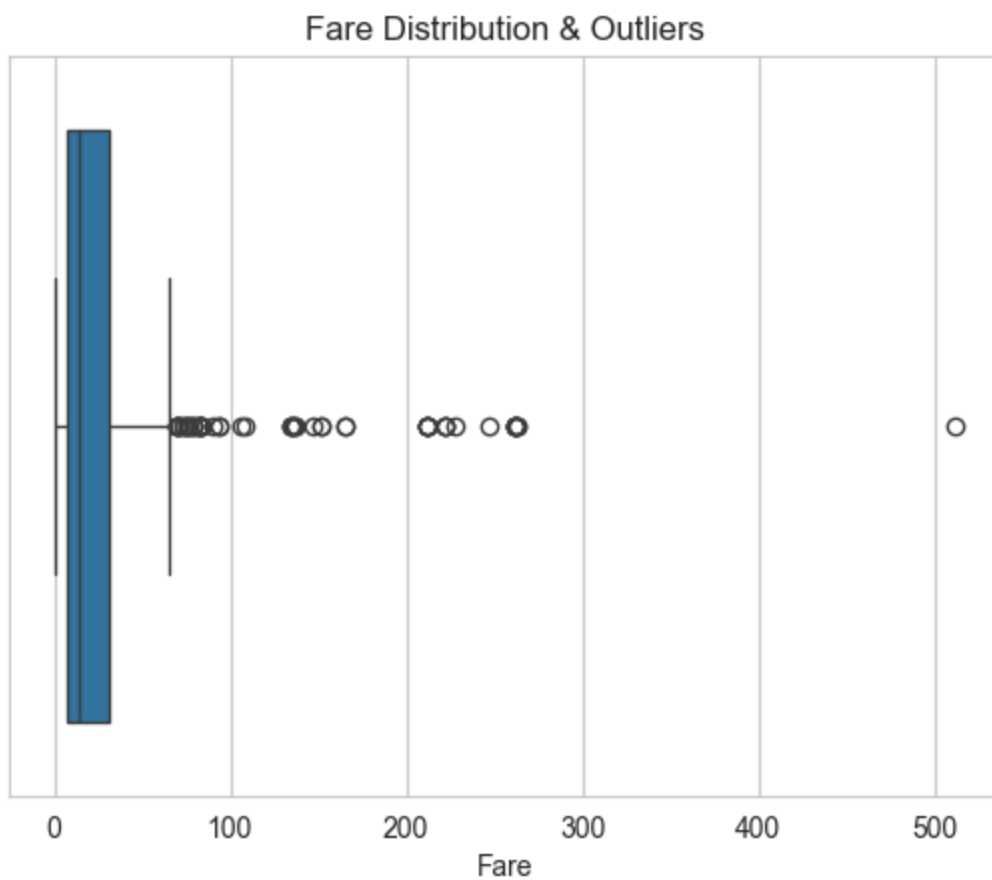
```
In [25]: #Q5. What is the Age Distribution?
sns.histplot(titanic['Age'], kde=True)
plt.title("Age Distribution")
plt.show()
```

```
In [26]: #Q6. Does Age Influence Survival?
sns.boxplot(x='Survived', y='Age', data=titanic)
plt.title("Age vs Survival")
plt.show()
```

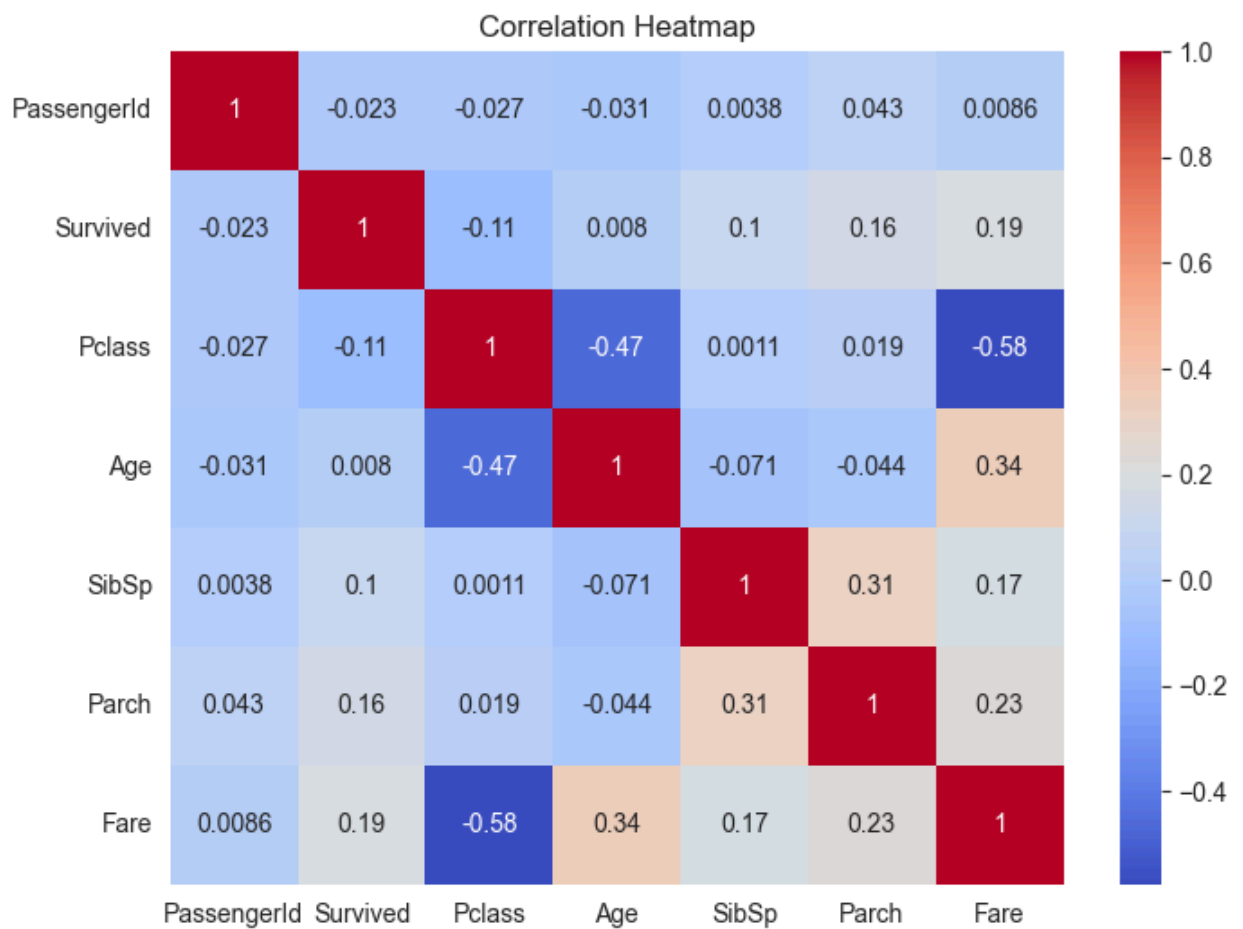


```
In [27]: #Q7. Fare Distribution & Outliers?
sns.boxplot(x=titanic['Fare'])
plt.title("Fare Distribution & Outliers")
plt.show()
```

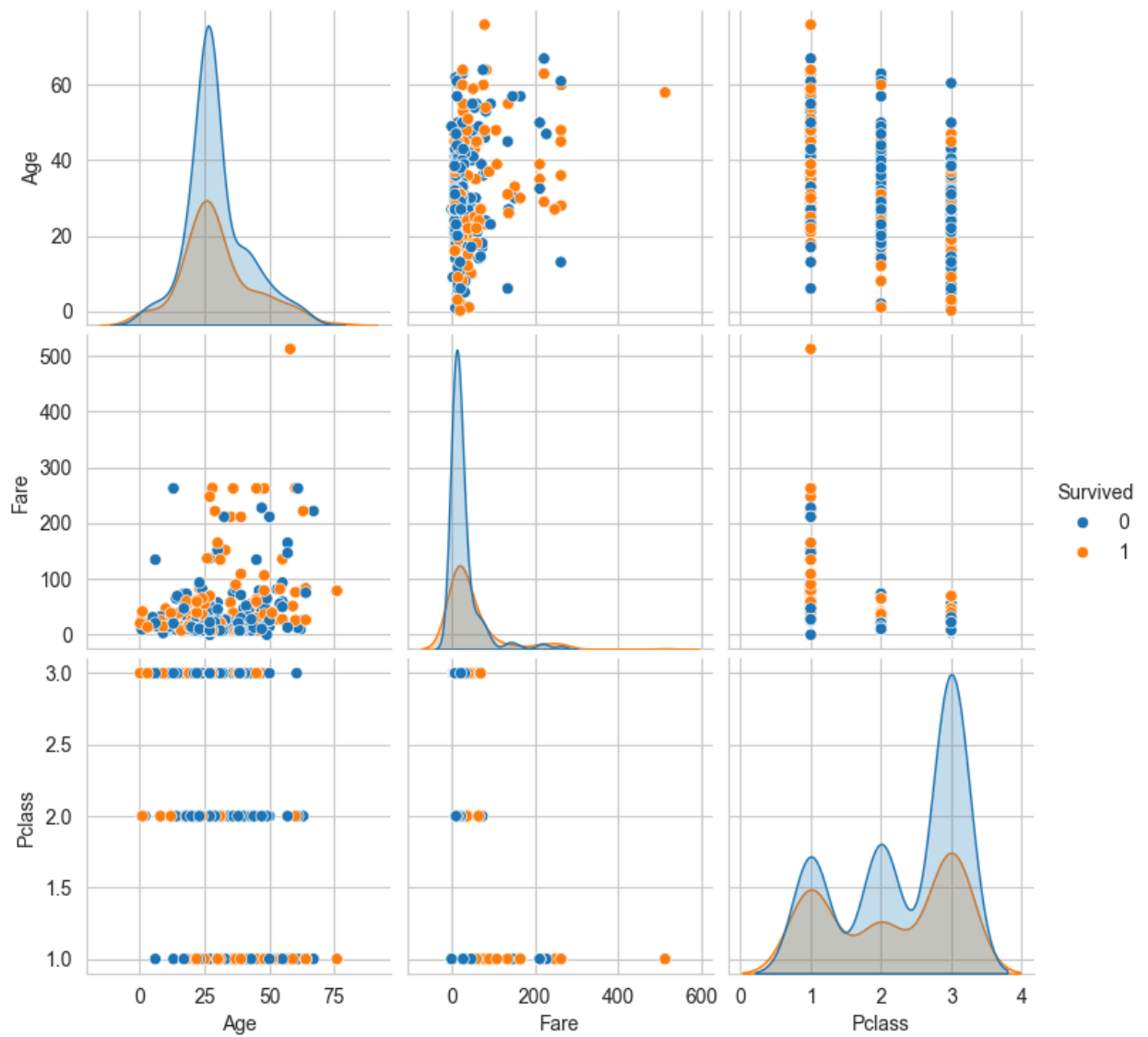


```
In [28]: #Q8. Correlation Analysis?
corr = titanic.corr(numeric_only=True)

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

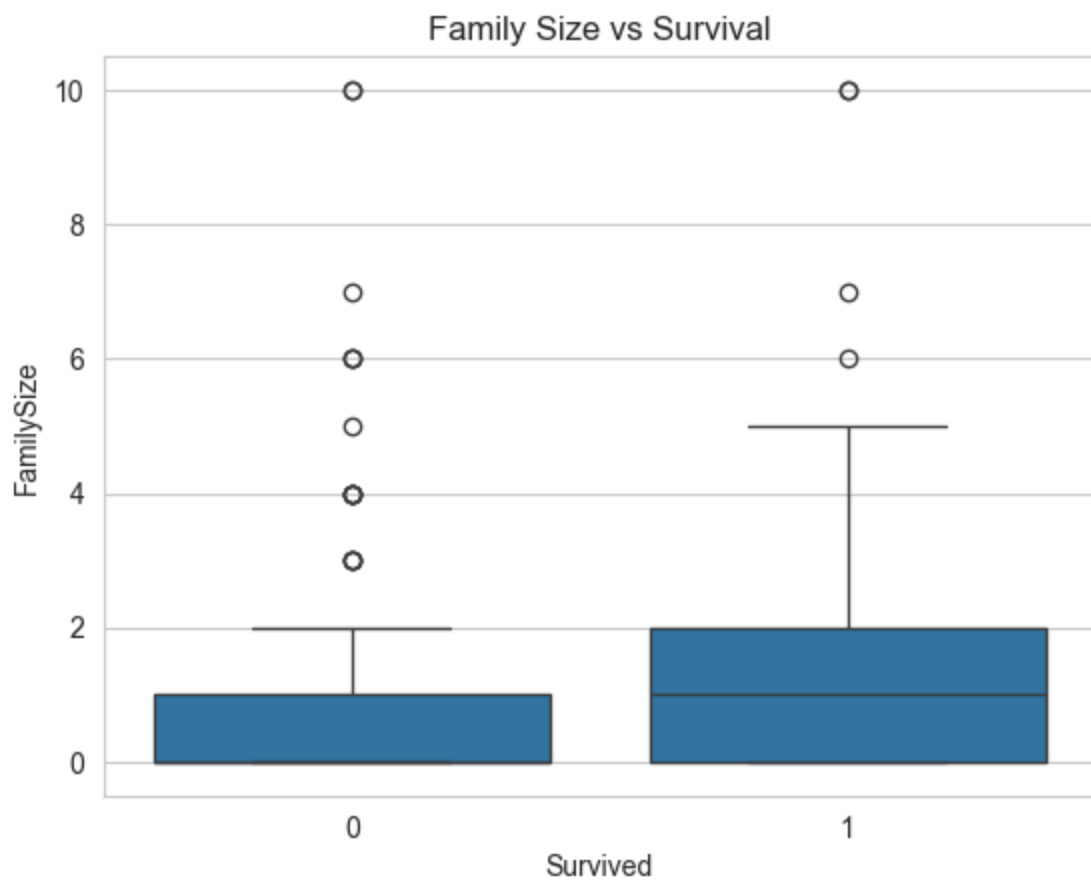


```
In [29]: #Q9. Pairwise Relationship Exploration?
sns.pairplot(titanic[['Age', 'Fare', 'Pclass', 'Survived']], hue='Survived')
plt.show()
```

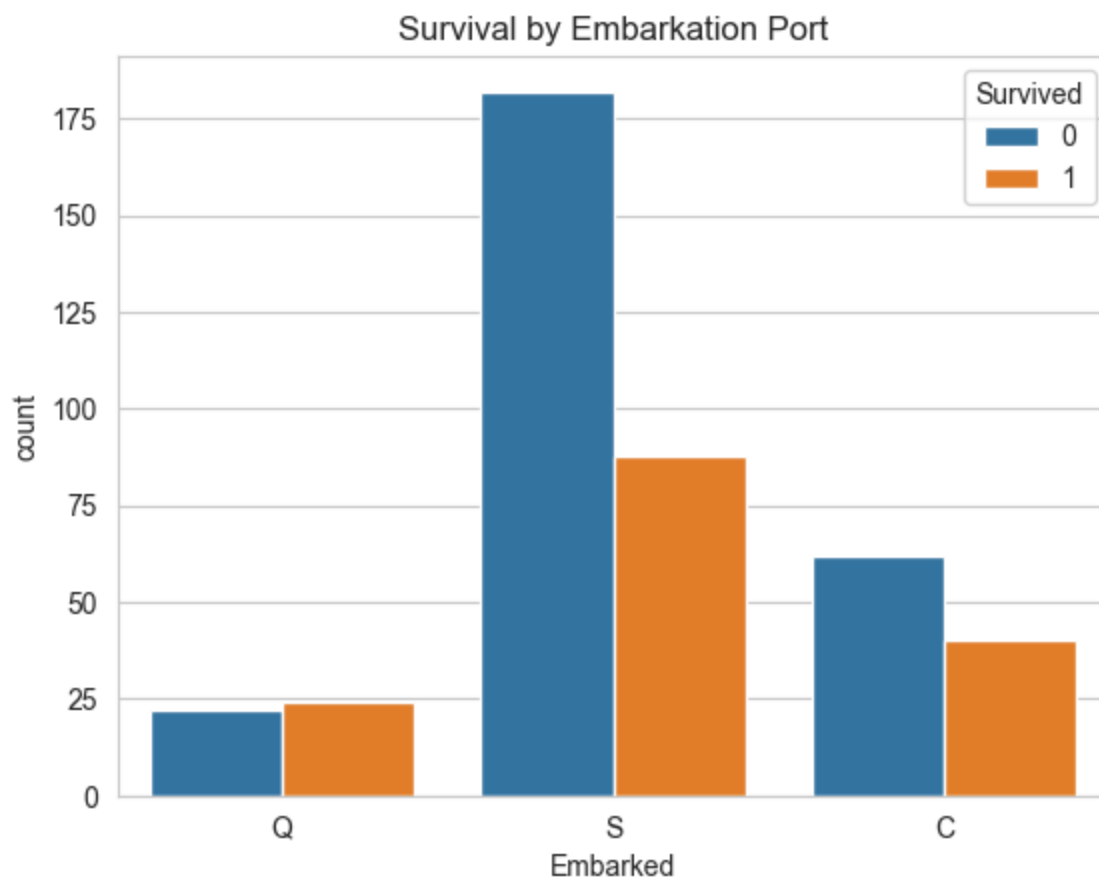


```
In [30]: #Q10. Does Family Size Impact Survival?
titanic['FamilySize'] = titanic['SibSp'] + titanic['Parch']

sns.boxplot(x='Survived', y='FamilySize', data=titanic)
plt.title("Family Size vs Survival")
plt.show()
```



```
In [31]: #Q11. Embarkation Port Impact?
sns.countplot(x='Embarked', hue='Survived', data=titanic)
plt.title("Survival by Embarkation Port")
plt.show()
```



```
In [32]: #Q12. Statistical Skewness Check?  
print(titanic.skew(numeric_only=True))
```

```
PassengerId    0.000000  
Survived       0.568991  
Pclass        -0.534170  
Age           0.660747  
SibSp         4.168337  
Parch         4.654462  
Fare          3.692299  
FamilySize    3.168542  
dtype: float64
```

```
In [33]: #Q13. Missing Value Analysis?  
print(titanic.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            0
SibSp           0
Parch           0
Ticket         0
Fare            0
Cabin           0
Embarked        0
FamilySize      0
dtype: int64
```

```
In [35]: #Q14. Survival Rate Analysis?
survival_rate = titanic['Survived'].mean()
print("Survival Probability:", survival_rate)
```

Survival Probability: 0.36363636363636365

```
In [36]: #Q15. Conditional Probability Analysis?
titanic.groupby('Sex')['Survived'].mean()
```

```
Out[36]: Sex
female    1.0
male      0.0
Name: Survived, dtype: float64
```

```
In [37]: #Q16. Survival by Passenger Class Analysis?
titanic.groupby('Pclass')['Survived'].mean()
```

```
Out[37]: Pclass
1      0.467290
2      0.322581
3      0.330275
Name: Survived, dtype: float64
```

```
In [38]: #Q17. Correlation Strength Analysis?
titanic.corr(numeric_only=True)['Survived'].sort_values(ascending=False)
```

```
Out[38]: Survived    1.000000
Fare              0.192036
FamilySize        0.161803
Parch             0.159120
SibSp             0.099943
Age               0.008035
PassengerId      -0.023245
Pclass           -0.108615
Name: Survived, dtype: float64
```

```
In [41]: #Q18. Family Size Effect Analysis?
titanic['FamilySize'] = titanic['SibSp'] + titanic['Parch']
titanic.groupby('FamilySize')['Survived'].mean()
```



```
Out[41]: FamilySize
0      0.268775
1      0.486486
2      0.526316
3      0.714286
4      0.285714
5      0.666667
6      0.250000
7      0.500000
10     0.500000
Name: Survived, dtype: float64
```

```
In [42]: #Q19. Multivariable Interaction Analysis?
pd.crosstab(titanic['Pclass'], titanic['Sex'], values=titanic['Survived'], agg
```

```
Out[42]:
```

	Sex female	male
Pclass		
1	1.0	0.0
2	1.0	0.0
3	1.0	0.0

```
In [ ]:
```